Introduction to Classification- Issues Regarding Classification and Prediction – Classification by Decision Tree Introduction – Bayesian Classification – Rule Based Classification – Classification by Back propagation – Support Vector Machines – Associative Classification – Lazy Learners – Other Classification Methods.

There are four main categories of Machine Learning algorithms: supervised, unsupervised, semi-supervised, and reinforcement learning.

Classification is a supervised machine learning method where the model tries to predict the correct label of a given input data. In classification, the model is fully trained using the training data, and then it is evaluated on test data before being used to perform prediction on new unseen data.

Before diving into the classification concept, we will first understand the difference between the two types of learners in classification: lazy and eager learners.

Learners in Classifications Algorithm

In machine learning, classification learners can also be classified as either “lazy” or “eager” learners.

Eager Learners: Eager Learners are also known as model-based learners, eager learners learn a model from the training data during the training phase and use this model to classify new instances at prediction time. It is more effective in high-dimensional spaces having large training datasets. Examples of eager learners include decision trees, random forests, and support vector machines.

Most machine learning algorithms are eager learners, and below are some examples:

- Logistic Regression.

- Support Vector Machine.
- Decision Trees.
- Artificial Neural Networks.

Lazy Learners: Lazy Learners are also known as instance-based learners, lazy learners do not learn a model during the training phase. Instead, they simply store the training data and use it to classify new instances at prediction time. It is very fast at prediction time because it does not require computations during the predictions. it is less effective in high-dimensional spaces or when the number of training instances is large.

- K-Nearest Neighbor.
- Case-based reasoning.

However, some algorithms, such as **BallTrees** and **KDTrees**, can be used to improve the prediction latency.

Classification Types

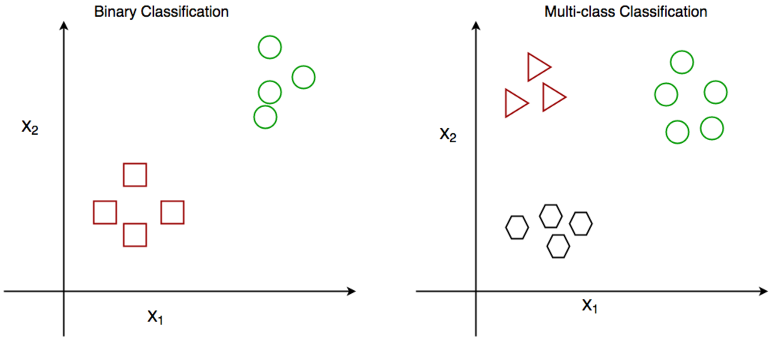
There are two main classification types in machine learning:

Binary Classification

In binary classification, the goal is to classify the input into one of two classes or categories. Example – On the basis of the given health conditions of a person, we have to determine whether the person has a certain disease or not.

Multiclass Classification

In multi-class classification, the goal is to classify the input into one of several classes or categories. For Example – On the basis of data about different species of flowers, we have to determine which specie our observation belongs to.



Binary vs Multi class classification

Other categories of classification involves:

Multi-Label Classification

In, **Multi-label Classification** the goal is to predict which of several labels a new data point belongs to. This is different from multiclass classification, where each data point can only belong to one class. For example, a multi-label classification algorithm could be used to classify images of animals as belonging to one or more of the categories cat, dog, bird, or fish.

Imbalanced Classification

In, **Imbalanced Classification** the goal is to predict whether a new data point belongs to a minority class, even though there are many more examples of the majority class. For example, a medical diagnosis algorithm could be used to predict whether a patient has a rare disease, even though there are many more patients with common diseases.

Training set—a subset to train a model. Test set—a subset to test the

trained model.

Classification Algorithms

There are various x. Some of them are :

Linear Classifiers

Linear(arranged in or extending along a straight or nearly straight line).models create a linear decision boundary between classes. They are simple and computationally efficient. Some of the linear **classification** models are as follows:

- Logistic Regression
- Support Vector Machines having kernel = ‘linear’
- Single-layer Perceptron
- Stochastic Gradient Descent (SGD) Classifier

Non-linear Classifiers

Non-linear models create a non-linear decision boundary between classes. They can capture more complex relationships between the input features and the target variable. Some of the non-linear **classification** models are as follows:

- K-Nearest Neighbours
- Kernel SVM
- Naive Bayes
- Decision Tree Classification
- Ensemble learning classifiers:
- Random Forests,
- AdaBoost,
- Bagging Classifier,
- Voting Classifier,
- ExtraTrees Classifier
- Multi-layer Artificial Neural Networks

The main difference between **continuous and categorical variables** is that continuous variables can be measured, while categorical variables describe objects:

- **Continuous variables**

Can be measured and represented as numbers for each unit in a sample. Examples include **age, height, weight, and temperature**. The value of a continuous variable is expressed as a mean, median, mode, range, standard deviation, and/or interquartile range.

- **Categorical variables**

Describe objects and are expressed as category frequencies in a sample. Examples include **gender, blood type, and hair color**. Categorical data can be further classified into nominal data, which has no inherent order or ranking, and ordinal data, which has an inherent order or ranking.

What is the Classification Algorithm?

The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups. Such as, **Yes or No, 0 or 1, Spam or Not Spam, cat or dog,** etc. Classes can be called as targets/labels or categories.

The output variable of Classification is a category, not a value, such as "Green or Blue", "fruit or animal", etc. Since the Classification algorithm is a Supervised learning technique, hence it takes labeled input data, which means it contains input with the corresponding output.

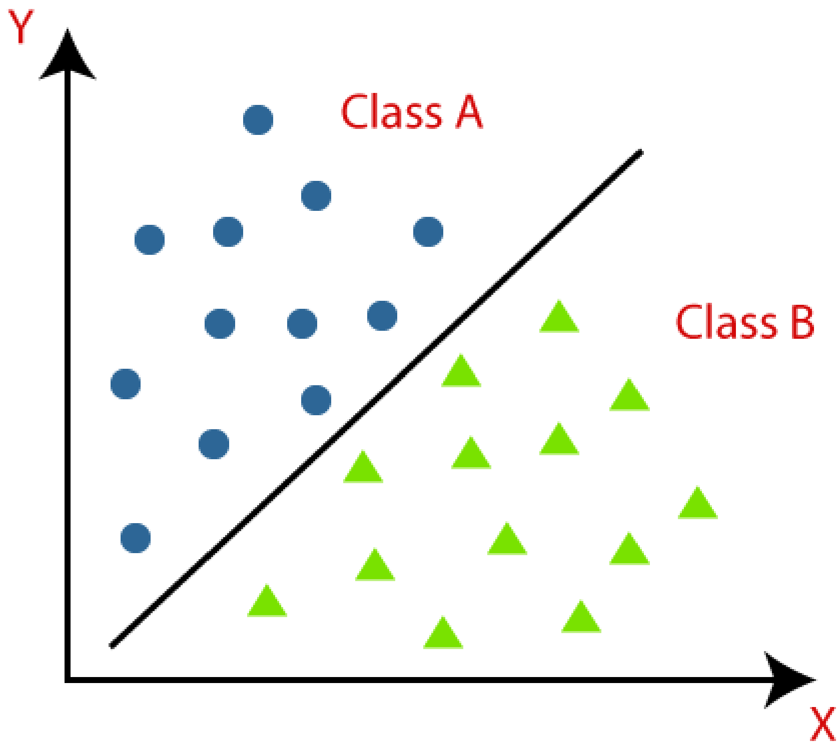
In classification algorithm, a discrete output function(y) is mapped to input variable(x).

y=f(x), where y = categorical output

The best example of an ML classification algorithm is **Email Spam Detector**.

The main goal of the Classification algorithm is to identify the category of a given dataset, and these algorithms are mainly used to predict the output for the categorical data.

Classification algorithms can be better understood using the below diagram. In the below diagram, there are two classes, class A and Class B. These classes have features that are similar to each other and dissimilar to other classes.



The algorithm which implements the classification on a dataset is known as a classifier. There are two types of Classifications:

- **Binary Classifier:** If the classification problem has only two possible outcomes, then it is called as Binary Classifier. **Examples:** YES or NO, MALE or FEMALE, SPAM or NOT SPAM, CAT or DOG, etc.
- **Multi-class Classifier:** If a classification problem has more than two outcomes, then it is called as Multi-class Classifier.

Example: Classifications of types of crops, Classification of types of music.

Evaluation metrics for classification m.odels in Machine Learning

Evaluating a classification model is an important step in machine learning, as it helps to assess the performance of the model on new, unseen data. There are several metrics and techniques that can be used to evaluate a classification model, depending on the specific problem and requirements. Here are some commonly used evaluation metrics:

- **Classification Accuracy:** The proportion of correctly classified instances over the total number of instances in the test set. It is a simple and intuitive metric but can be misleading in imbalanced datasets where the majority class dominates the accuracy score.
- **Confusion matrix:** A table that shows the number of true positives, true negatives, false positives, and false negatives for each class, which can be used to calculate various evaluation metrics.
- **Precision and Recall:** Precision measures the proportion of true positives over the total number of predicted positives, while recall measures the proportion of true positives over the total number of actual positives. These metrics are useful in scenarios where one class is more important than the other,
- **F1-Score:** The harmonic mean of precision and recall, calculated as 2 x (precision x recall) / (precision + recall). It is a useful metric for imbalanced datasets where both precision and recall are important.
- **ROC curve and AUC:** The Receiver Operating Characteristic (ROC) curve is a plot of the true positive rate (recall) against the false positive rate (1-specificity) for different threshold values of the classifier's decision function. The Area Under the Curve (AUC) measures the overall

performance of the classifier, with values ranging from 0.5 (random guessing) to 1 (perfect classification).

Steps in building the classification Model

The process of building a classification model typically involves the following steps:

Data Collection:

The first step in building a classification model is data collection. In this step, the data relevant to the problem at hand is collected. The data should be representative of the problem and should contain all the necessary attributes and labels needed for classification. The data can be collected from various sources, such as surveys, questionnaires, websites, and databases.

Data Preprocessing:

The second step in building a classification model is data preprocessing. The collected data needs to be preprocessed to ensure its quality. This involves handling missing values, dealing with outliers, and transforming the data into a format suitable for analysis. Data preprocessing also involves converting the data into numerical form, as most classification algorithms require numerical input.

Handling Missing Values: Missing values in the dataset can be handled by replacing them with the mean, median, or mode of the corresponding feature or by removing the entire record.

Dealing with Outliers: Outliers in the dataset can be detected using various statistical techniques such as z-score analysis, boxplots, and scatterplots. Outliers can be removed from the dataset or replaced with the mean, median, or mode of the corresponding feature.

Data Transformation: Data transformation involves scaling or normalizing the data to bring it into a common scale. This is done to ensure that all features have the same level of importance in the analysis.

Feature Selection:

The third step in building a classification model is feature selection. Feature selection involves identifying the most relevant attributes in the dataset for classification. This can be done using various techniques, such as correlation analysis, information gain, and principal component analysis.

Correlation Analysis: Correlation analysis involves identifying the correlation between the features in the dataset. Features that are highly correlated with each other can be removed as they do not provide additional information for classification.

Information Gain: Information gain is a measure of the amount of information that a feature provides for classification. Features with high information gain are selected for classification.

Principal Component Analysis:

Principal Component Analysis (PCA) is a technique used to reduce the dimensionality of the dataset. PCA identifies the most important features in the dataset and removes the redundant ones.

Model Selection:

The fourth step in **building a classification model** is model selection. Model selection involves selecting the appropriate classification algorithm for the problem at hand. There are several algorithms available, such as decision trees, support vector machines, and neural networks.

Decision Trees:

Decision trees are a simple yet powerful classification algorithm. They divide the dataset into smaller subsets based on the values of the features and construct a tree-like model that can be used for classification.

Support Vector Machines:

Support Vector Machines (SVMs) are a popular classification algorithm used for both linear and nonlinear classification problems. SVMs are based on the concept of maximum margin, which involves finding the hyperplane that maximizes the distance between the two classes.

Neural Networks:

Neural Networks are a powerful classification algorithm that can learn complex patterns in the data. They are inspired by the structure of the human brain and consist of multiple layers of interconnected nodes.

Model Training:

The fifth step in building a classification model is **model training**. Model training involves using the selected classification algorithm to learn the patterns in the data. The data is divided into a **training set and a validation set**. The model is trained using the training set, and its performance is evaluated on the validation set.

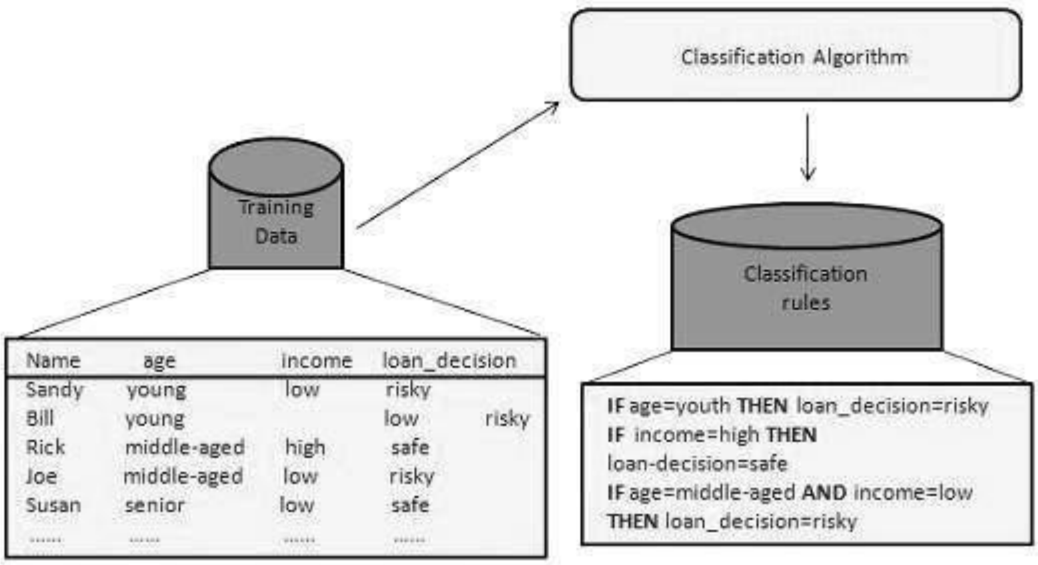
Model Evaluation:

The sixth step in building a classification model is model evaluation. Model evaluation involves assessing the performance of the trained model on a test set.

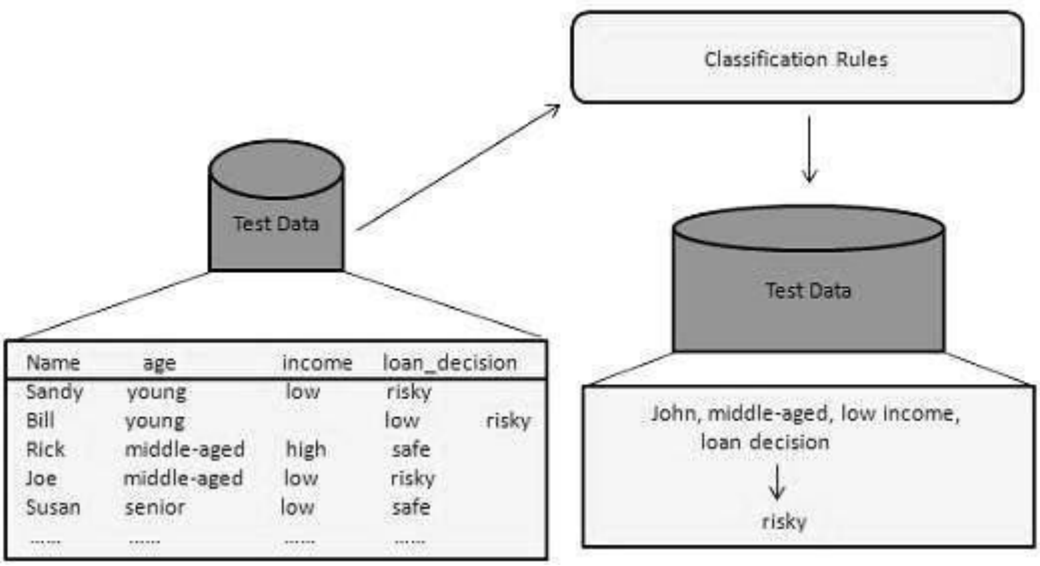
How Does Classification Works?

The Data Classification process includes two steps

Learning Step (Training Phase): Construction of Classification Model Different Algorithms are used to build a classifier by making the model learn using the training set available. The model has to be trained for the prediction of accurate results.



Classification Step: Model used to predict class labels and testing the constructed model on test data and hence estimate the accuracy of the classification rules.



Associated Tools and Languages: Used to mine/ extract useful information from raw data.

- **Main Languages used**: R, Python, SQL
- **Major Tools used**: RapidMiner, Orange, KNIME, Spark, Weka
- **Libraries used**: Jupyter, NumPy, Matplotlib, Pandas, ScikitLearn, NLTK, TensorFlow, Seaborn, Basemap, etc.

Real–Life Examples :

- **Market Basket Analysis:**

  It is a modeling technique that has been associated with frequent transactions of buying some combination of items. **Example**: Amazon and many other Retailers use this technique. While viewing some products, certain suggestions for the commodities are shown that some people have bought in the past.
- **Weather Forecasting:**

  Changing Patterns in weather conditions needs to be observed based on parameters such as temperature, humidity, wind direction. This keen observation also requires the use of previous records in order to predict it accurately.

Advantages:

- Mining Based Methods are cost-effective and efficient
- Helps in identifying criminal suspects
- Helps in predicting the risk of diseases
- Helps Banks and Financial Institutions to identify defaulters so that they may approve Cards, Loan, etc.

Disadvantages:

Privacy: When the data is either are chances that a company may give some information about their customers to other vendors or use this information for

their profit.
Accuracy Problem: Selection of Accurate model must be there in order to get the best accuracy and result.

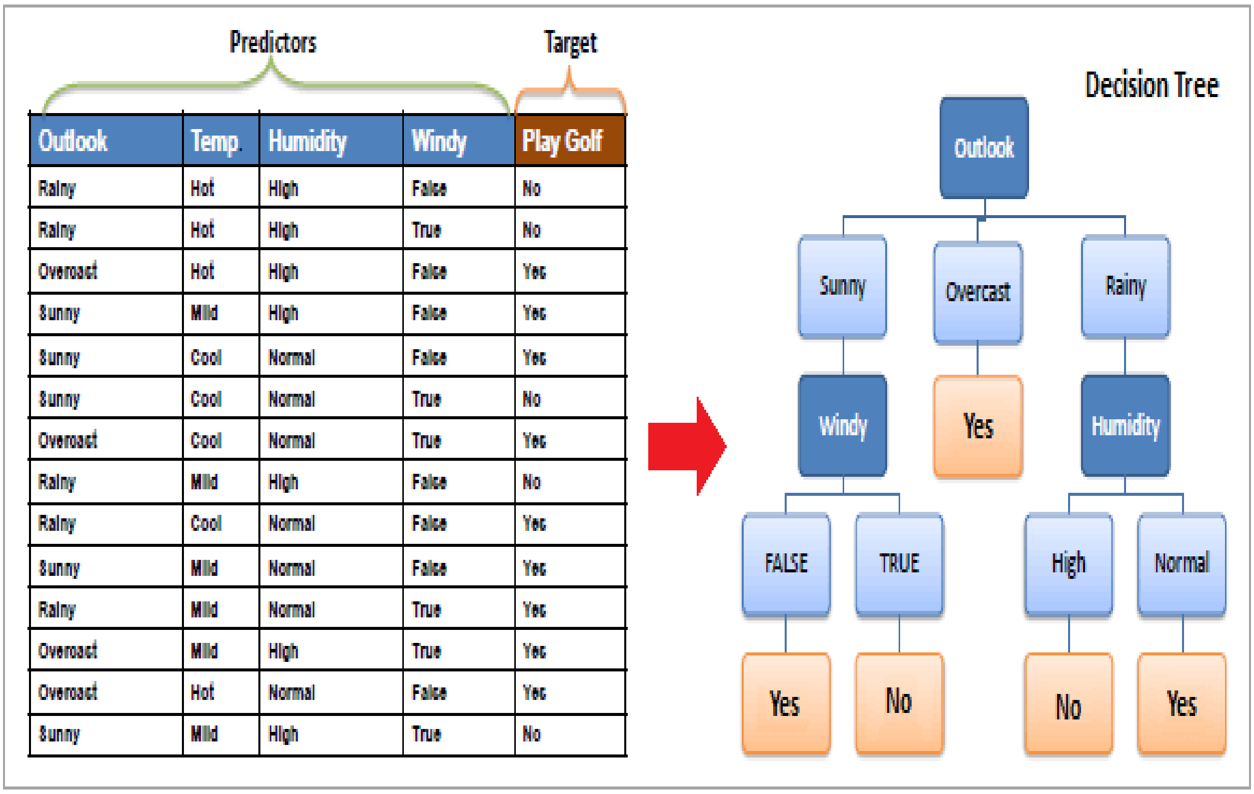
Decision Tree Algorithm

- Decision Tree Mining is a type of data mining technique that is used to build Classification Models.
- It builds classification models in the form of a tree-like structure, just like its name.
- This type of mining belongs to supervised class learning (the target result is already known).
- Decision trees can be used for both categorical and numerical data. The categorical data represent gender, marital status, etc. while the numerical data represent age, temperature, etc.
- It is used to create data models that will predict class labels or values for the decision-making process.

Decision tree

A decision tree is a structure that includes a root node, branches, and leaf nodes. Each internal node denotes a test on an attribute, each branch denotes the outcome of a test, and each leaf node holds a class label. The topmost node in the tree is the **root node**.

Example:



Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

How Does A Decision Tree Work?

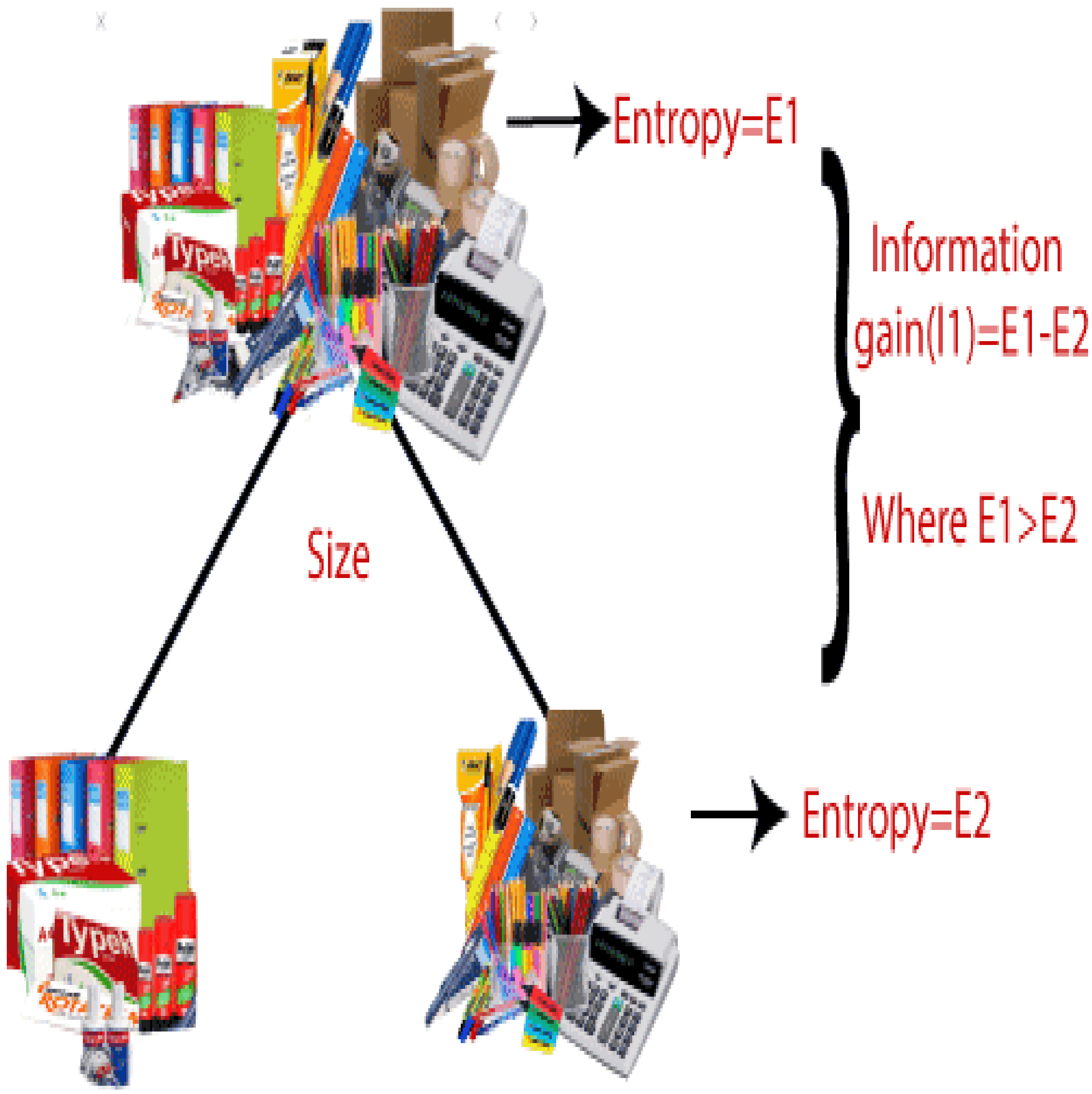
A decision tree is a supervised learning algorithm that works for both discrete and continuous variables. It splits the dataset into subsets on the basis of the most significant attribute in the dataset. How the decision tree identifies this attribute and how this splitting is done is decided by the algorithms.

The most significant predictor is designated as the root node, splitting is done to form sub-nodes called decision nodes, and the nodes which do not split further are terminal or leaf nodes.

It follows a top-down approach as the top region presents all the observations at a single place which splits into two or more branches that further split. This approach is also called a greedy approach as it only considers the current node between the worked on without focusing on the future nodes.

The decision tree algorithms will continue running until a stop criterion such as the minimum number of observations etc. is reached. Once a decision tree is built, many nodes may represent outliers or noisy data. **Tree pruning** method is applied to remove unwanted data. This, in turn, improves the accuracy of the classification model.

Key factors:



Entropy:

Entropy refers to a common way to measure impurity. In the decision tree, it measures the impurity in data sets.

Information Gain:

Information Gain refers to the decline in entropy after the dataset is split. It is also called Entropy Reduction. Building a decision tree is all about discovering attributes that return the highest data gain.

Important terminology

Root Node:

This attribute is used for dividing the data into two or more sets. The feature attribute in this node is selected based on Attribute Selection Techniques.

Branch or Sub-Tree:

A part of the entire decision tree is called a branch or sub-tree.

Splitting:

Dividing a node into two or more sub-nodes based on if-else conditions.

Decision Node:

After splitting the sub-nodes into further sub-nodes, then it is called the decision node.

Leaf or Terminal Node:

This is the end of the decision tree where it cannot be split into further sub-nodes.

Pruning:

Removing a sub-node from the tree is called pruning.

Some of the decision tree algorithms include **Hunt's Algorithm, ID3, CD4.5, and CART.**

Decision Tree Induction

Decision tree induction is the method of learning the decision trees from the training set. The training set consists of attributes and class labels. **Applications of decision tree induction include astronomy, financial analysis, medical diagnosis, manufacturing, and production**.

A decision tree is a flowchart tree-like structure that is made from training set tuples. The dataset is broken down into smaller subsets and is present in the form of nodes of a tree. The tree structure has a root node, internal nodes or decision nodes, leaf node, and branches.

The root node is the topmost node. It represents the best attribute selected for classification. Internal nodes of the decision nodes represent a test of an attribute of the dataset leaf node or terminal node which represents the classification or decision label. The branches show the outcome of the test performed.

Some decision trees only have binary nodes that means exactly two branches of a node, while some decision trees are non-binary.

Advantages of the Decision Tree

- It is simple to understand as it follows the same process which a human follow while making any decision in real-life.
- It can be very useful for solving decision-related problems.
- It helps to think about all the possible outcomes for a problem.
- There is less requirement of data cleaning compared to other algorithms.

Disadvantages of the Decision Tree

- The decision tree contains lots of layers, which makes it complex.
- It may have an overfitting issue, which can be resolved using the **Random Forest algorithm.**
- For more class labels, the computational complexity of the decision tree may increase.

Rule based Classification

What is Rule-based Classification in Data Mining?

Rule-based classification in data mining is a technique in which class decisions are taken based on various “if...then… else” rules. Thus, we define it as a classification type governed by a set of IF-THEN rules. The rule was written aas IF-THEN rule as:

“IF condition THEN conclusion.”

IF-THEN Rule

To define the IF-THEN rule, we can split it into two parts:

- **Rule Antecedent:** This is the “if condition” part of the rule. This part is present in the LHS (Left Hand Side). The antecedent can have one or more attributes as conditions, with logic AND operator.

- **Rule Consequent:** This is present in the rule's RHS (Right Hand Side). The rule consequent consists of the class prediction.

Assessment of Rule

In rule-based classification in data mining, there are two factors based on which we can access the rules. These are:

- **Coverage of Rule:**

  The **fraction of the records which satisfy the antecedent conditions of a particular rule is called the coverage of that rule**. We can calculate this by dividing the number of records satisfying the rule(n1) by the total number of records(n). Coverage(R)=n1/n

- **Accuracy of a rule:**

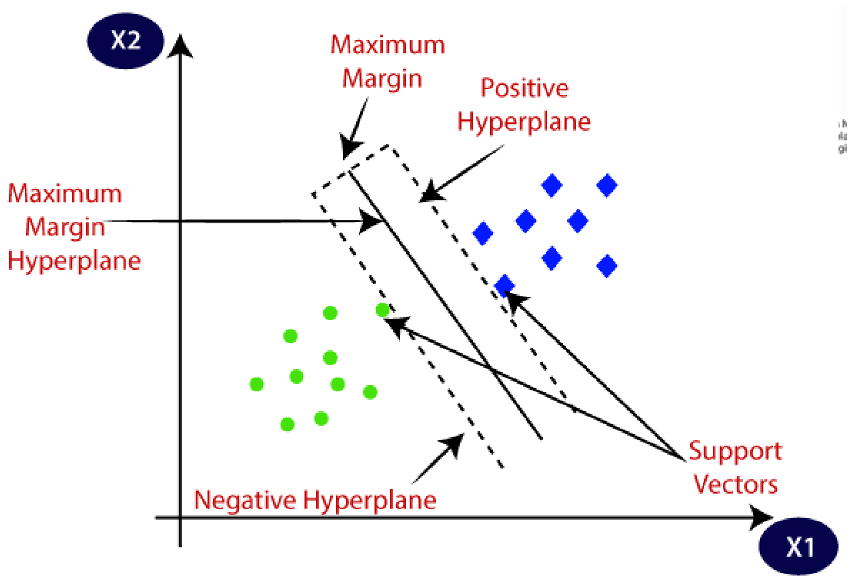
  The fraction of the records that satisfy the antecedent conditions and meet the consequent values of a rule is called the accuracy of that rule.
  We can calculate this by dividing **the number of records satisfying the consequent values(n2)** by the **number of records satisfying the rule(n1).**
  Accuracy(R) = n2/n1

SUPPORT VECTOR MACHINE

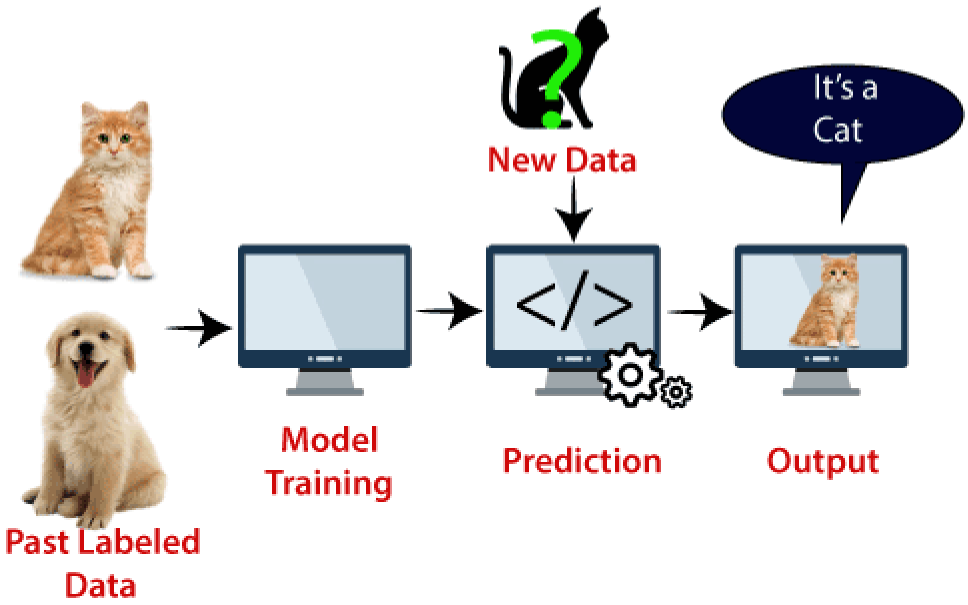
Support vector machines so called as SVM is a **supervised learning algorithm** which can be used for classification and regression problems as support vector classification (SVC) and support vector regression (SVR). It is more preferred for classification. It is used for smaller dataset as it takes too long to process. **SVM is a supervised classification method that separates data using hyperplanes**. SVM **finds a hyper-plane** that creates a boundary between the types of data.

The goal of the SVM algorithm is **to create the best line or decision boundary** that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a **hyperplane**.

SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



Example: Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm. We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature. So as support vector creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog. On the basis of the support vectors, it will classify it as a cat. Consider the below diagram:



Types of SVM

SVM can be of two types:

Linear SVM: Linear SVM is used for linearly separable data, which means **if a dataset can be classified into two classes by using a single straight line**, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.

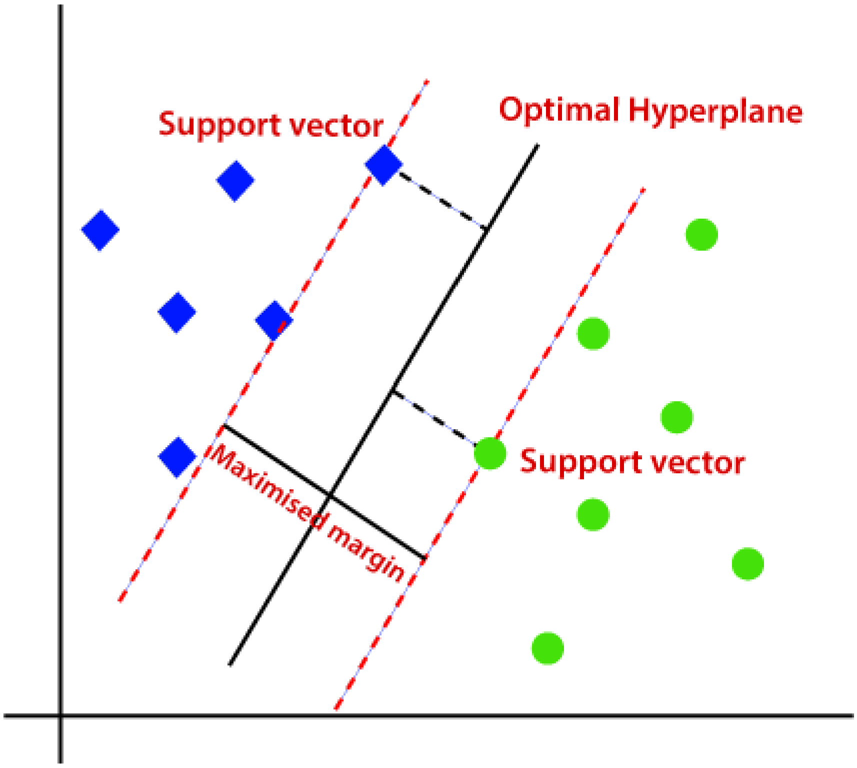
Non-linear SVM: Non-Linear SVM is used for non-linearly separated data, which means **if a dataset cannot be classified by using a straight line**, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Hyperplane and Support Vectors in the SVM algorithm:

Hyperplane: There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points. **This best boundary is known as the hyperplane of SVM.**

Support Vectors:

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.



Key Concepts:*

1. Hyperplane: Decision boundary separating classes.

2. Margin: Distance between hyperplane and nearest data points.

3. Support Vectors: Data points closest to the hyperplane.

4. Kernel Function: Maps data to higher-dimensional space.

How SVM Works:*

1. Data Preprocessing: Normalize and transform data.

2. Choose Kernel: Select suitable kernel function (e.g., linear, polynomial, radial basis function).

3. Train Model: Find optimal hyperplane maximizing margin.

4. Classification: Predict class labels for new data.

Advantages:

1. High accuracy

2. Robust to noise

3. Effective in high-dimensional spaces

4. Flexible kernel choices

Disadvantages:

1. Computational complexity

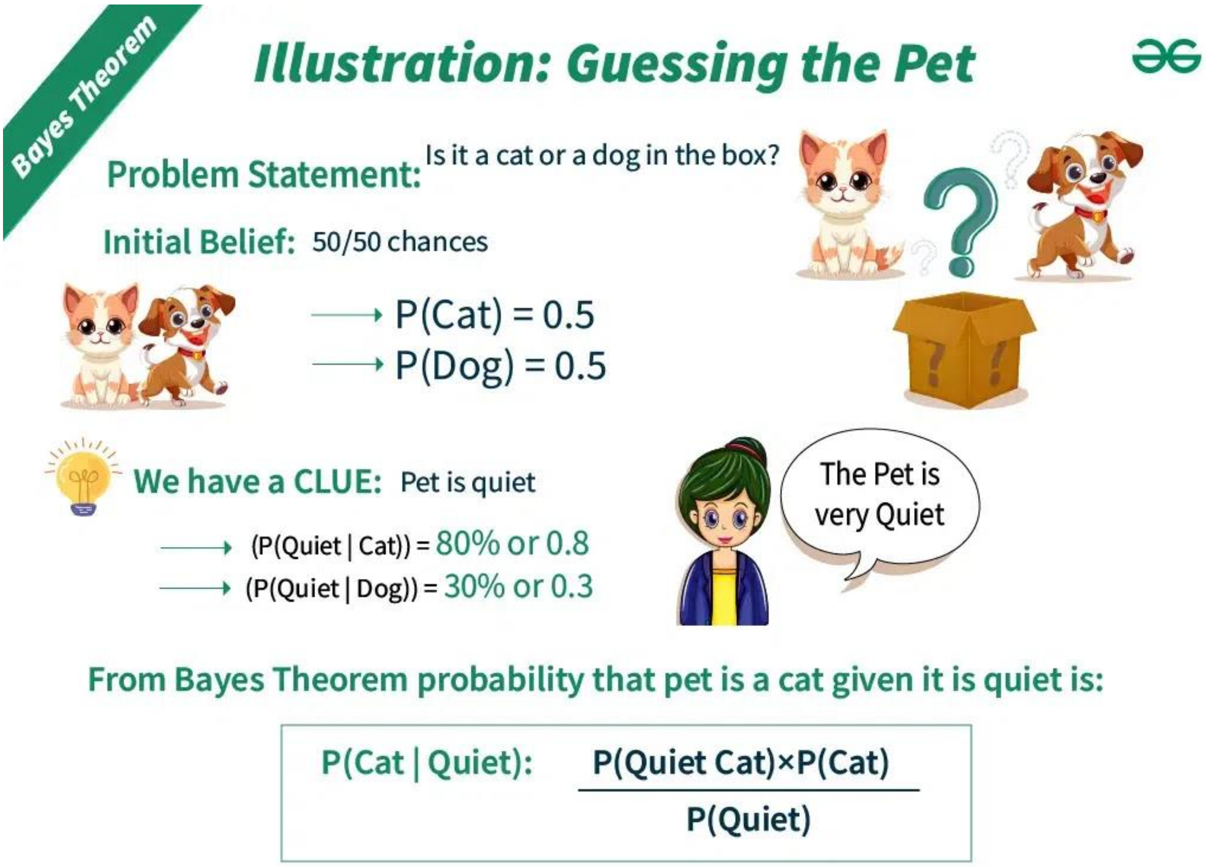
2. Difficulty in selecting optimal kernel

3. Overfitting risk

Bayes’ Theorem

Bayes' Theorem is used to determine the conditional probability of an event. It was named after an English statistician, Thomas Bayes who discovered this formula in 1763. Bayes Theorem is a very important theorem in mathematics,

It is used to find the probability of an event, based on prior knowledge of conditions that might be related to that event.



Bayes Theorem Formula

For any two events A and B, then the formula for the Bayes theorem is given by: (the image given below gives the Bayes' theorem formula)

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Bayes' Theorem Formula

where,

- P(A) and P(B) are the probabilities of events A and B also P(B) is never equal to zero.
- P(A|B) is the probability of event A when event B happens
- P(B|A) is the probability of event B when A happens

Bayesian Classification

Bayesian classification uses Bayes theorem to predict the occurrence of any event. Bayesian classifiers are the statistical classifiers with the Bayesian probability understandings.

Types of Bayesian Classifiers

1. Naive Bayes: Assumes independence between features.

2. Bayesian Networks: Models relationships between features.

3.Hierarchical Bayes: Models hierarchical relationships.

Advantages

1.Handling uncertainty: Bayesian classification naturally handles uncertainty.

2. Flexibility: Can incorporate prior knowledge.

3. Interpretability: Provides probability estimates.

Disadvantages

1. Computational complexity: Can be computationally expensive.

2. Overfitting: May overfit with complex models.

Applications

1. Spam filtering

2. Image classification

3. Sentiment analysis

4. Medical diagnosis

BACK PROPOGATION CLASSIFICATION

Backpropagation is an algorithm that backpropagates the errors from the output nodes to the input nodes. Therefore, it is simply referred to as the backward propagation of errors.

Backpropagation, short for "backward propagation of errors," is a fundamental algorithm used for training artificial neural networks.

Neural networks are an information processing paradigm inspired by the human nervous system. Just like in the human nervous system, we have biological neurons in the same way in neural networks we have artificial neurons, artificial neurons are mathematical functions derived from biological neurons. The human brain is estimated to have about 10 billion neurons, each connected to an average of 10,000 other neurons. Each neuron receives a signal through a synapse, which controls the effect of the signconcerning on the neuron.

Working of Backpropagation:

Neural networks use supervised learning to generate output vectors from input vectors that the network operates on. It Compares generated output to the desired output and generates an error report if the result does not match the generated output vector. Then it adjusts the weights according to the bug report to get your desired output.

Backpropagation Algorithm:

Step 1: Inputs X, arrive through the preconnected path.

Step 2: The input is modeled using true weights W. Weights are usually chosen randomly.

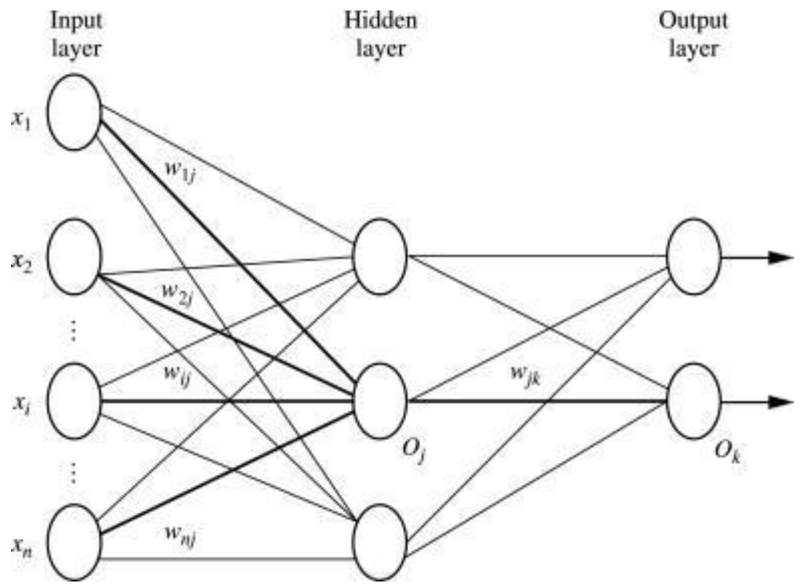
Step 3: Calculate the output of each neuron from the input layer to the hidden layer to the output layer.

Step 4: Calculate the error in the outputs

Backpropagation Error= Actual Output – Desired Output

Step 5: From the output layer, go back to the hidden layer to adjust the weights to reduce the error.

Step 6: Repeat the process until the desired output is achieved.

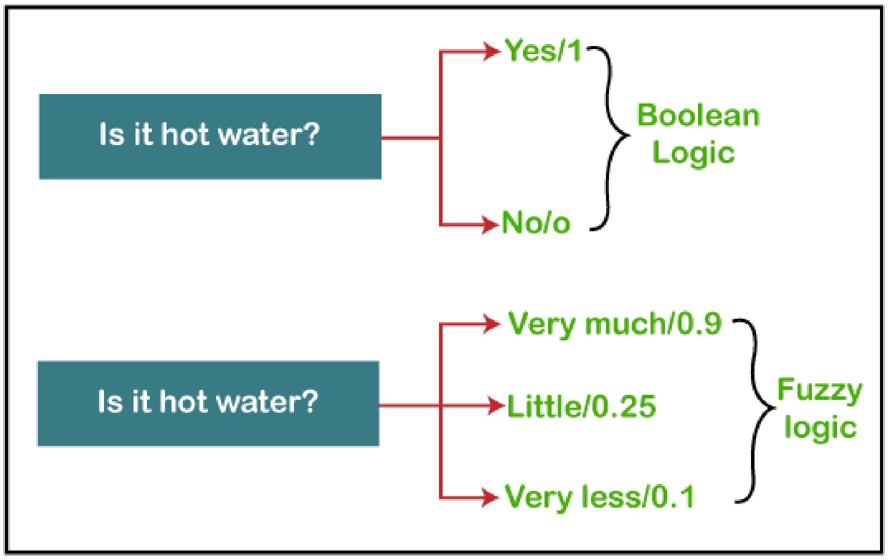


OTHER CLASSIFICATION TECHNIQUES

Fuzzy Logic

The 'Fuzzy' word means the things that are not clear or are vague. Sometimes, we cannot decide in real life that the given problem or statement is either true or false. At that time, this concept provides many values between the true and false and gives the flexibility to find the best solution to that problem.

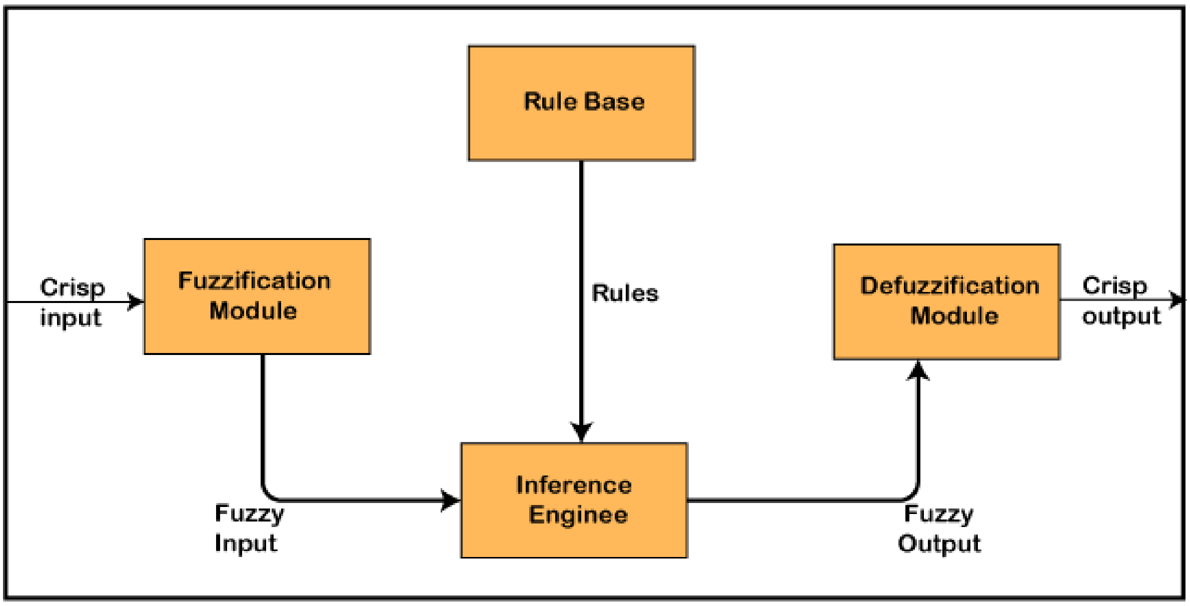
Example of Fuzzy Logic as comparing to Boolean Logic



- Fuzzy logic contains the multiple logical values and these values are the truth values of a variable or problem between 0 and 1.
- This concept provides the possibilities which are not given by computers, but similar to the range of possibilities generated by humans.
- In the Boolean system, only two possibilities (0 and 1) exist, where 1 denotes the absolute truth value and 0 denotes the absolute false value. But in the fuzzy system, there are multiple possibilities present between the 0 and 1, which are partially false and partially true.

Architecture of a Fuzzy Logic System

- In the architecture of the Fuzzy Logic system, each component plays an important role. The architecture consists of the different four components which are given below.
  - Rule Base
  - Fuzzification
  - Inference Engine
  - Defuzzification



1. Rule Base

Rule Base is a component used for storing the set of rules and the If-Then conditions given by the experts are used for controlling the decision-making systems.

2. Fuzzification

Fuzzification is a module or component for transforming the system inputs, i.e., it converts the crisp number into fuzzy steps. The crisp numbers are those inputs which are measured by the sensors and then fuzzification passed them into the control systems for further processing.

3. Inference Engine

This component is a main component in any Fuzzy Logic system (FLS), because all the information is processed in the Inference Engine. It allows users to find the matching degree between the current fuzzy input and the rules.

4. Defuzzification

Defuzzification is a module or component, which takes the fuzzy set inputs generated by the Inference Engine, and then transforms them into a crisp value. It is the last step in the process of a fuzzy logic system.

UNIT II

SECTION A

1. Define Classification.
2. What are eager learners.
3. what are lazy learners.
4. What is continuous variable.
5. Define Binary Classification.
6. Name some non-linear classification models.
7. Name the steps involved in building a classification model.
8. Define feature selection.
9. Define SVM.
10. Define Information Gain.
11. Name some of the Advantages of the Decision Tree.
12. What is rule based classification in data mining.
13. Define Support Vectors.
14. Define Bayesian classification.
15. Define fuzzy logic.

SECTION B

1. Briefly explain Learners in Classifications Algorithm.
2. Write short notes on Classification Types.
3. Explain the types of classifiers algorithms.

4. Explain the steps involved in building a classification model.
5. Write short notes Evaluation metrics for classification models.
6. Briefly explain various techniques for feature selection.
7. Explain how does classification work.
8. Discuss the advantages and disadvantages of decision tree.
9. Explain Rule based Classification in detail.
10. Write short on Bayes' Theorem.

SECTION C

1. Describe decision tree algorithm and explain how it works.
2. Explain in detail about back propogation classification.
3. Explain in detail about Support Vector Machine.
4. Explain in detail about Fuzzy Logic classification techniques.
5. Explain in detail about Bayesian Classification.